

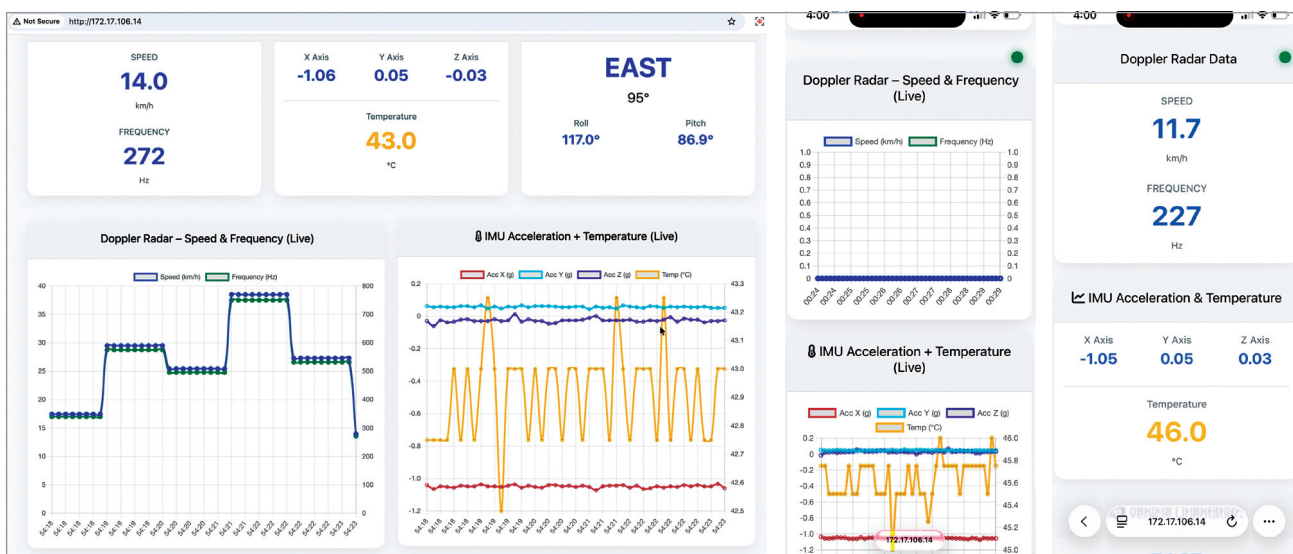


Fig. 10: CAN bus radar data on real-time dashboard

is intended for testing devices that receive the data over the CAN bus and display it on the serial monitor.

CAN bus radar code. In the CAN bus radar code, the first step is to configure the SSID and password for connecting to a Wi-Fi network. The device initially attempts to connect to the configured SSID. If the connection fails, it automatically switches to access point (AP) mode, where it acts as a router and creates its own Wi-Fi network. A phone or laptop can then be connected to this network.

Next, CAN bus IDs are assigned for transmitting different types of data. In this design, the following CAN IDs are used:

0x100 for IMU data (accelerometer and temperature)

0x200 for Doppler radar data (speed and frequency), and

0x300 for events and orientation

After this, a web-based dashboard is developed using HTML, CSS, JavaScript, and Chart.js for graphical visualisation. The system processes data from the radar, temperature sensor, and IMU, and implements an accident and collision-detection algorithm based on these inputs. All processed data is then transmitted over the TWAI CAN bus as well as displayed on the web dashboard. Fig. 7 shows the CAN bus ID settings and SSID/password configuration.

CAN bus receiver code. Next, the CAN bus receiver code is created. In this code, the system attempts to receive data from the previously defined CAN bus IDs and display it on the serial port. Fig. 8 shows the CAN bus receiver code snippet.

Construction and testing

After uploading the source code to the IndusBoard and connecting the HB100 radar and other components as per the circuit diagram (see Fig. 5 and Fig. 6), the system can be assembled into a custom-designed plastic enclosure. The radar can operate from behind the enclosure because its waves can penetrate a thin plastic layer. This makes the system more resistant to water and dust when enclosed.

Fig. 9 shows the radar assembly and its mounting inside the enclosure. Alternatively, the radar can be mounted directly on the car bonnet for improved durability and better

exposure.

First, design and 3D-print the enclosure for the radar module. Then, securely fix all components inside the enclosure. Finally, connect the radar module to the vehicle's CAN bus and power supply using the appropriate connector. Provide a regulated 5V supply to power the radar, then close the enclosure (see Fig. 9).

For testing, connect the CAN bus radar to the CAN bus receiver as per the circuit diagram. Then, open the serial port to view the data received from the radar.

To monitor the data transmitted by the CAN bus radar on the web dashboard, connect a phone or laptop to the same Wi-Fi network as the radar. Alternatively, if the device is in AP mode, connect to the radar's Wi-Fi network using the password defined in the code.

Once connected, open a web browser and enter the device's IP address. For example, in AP mode, the default IP address is 192.168.4.1. The dashboard then displays real-time data and graphical plots on a laptop or smartphone. For more details, see Fig. 10, which shows the CAN bus radar data on the real-time dashboard. **EFY**

Ashwini Kumar Sinha, an IoT and AI enthusiast, is Tech Journalist at EFY.

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